FORMAL MANUSCRIPT SUBMISSION COVER LETTER

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Publishing Model: Traditional Subscription Publishing Model (No Open Access Fee)

To: Editor-in-Chief Prof. Mohamed A. Abdou & Editorial Board
Fusion Engineering and Design

Dear Editor-in-Chief and Editorial Board Members,

I am pleased to submit our original research manuscript entitled **"Evaluating Vacuum Vessel Structural Reserves Under Asymmetric Halo Currents Using 3D Anisotropic Tsai-Wu Failure Envelopes"** for consideration as a Regular Research Article in *Fusion Engineering and Design*.

1. Scientific Scope & Core Novelty
This work presents a rigorous, self-contained mathematical and computational physics framework addressing critical fusion engineering bottlenecks. All field equations are established over real algebraic geometry with an explicit localized energy-mass density positivity constraint (T■■ ≥ 0). Under asymmetric n = 1 rotating halo currents (1.8 MA, 50–200 Hz), peak dynamic amplification (DAF = 1.85) yields a stress state (-158.5 MPa, -110.0 MPa, τ■■ = 180 MPa) with a Tsai-Wu failure metric f_TW = 0.244 ≤ 1.0 and Strength Reserve Ratio SRR = 1.61, proving structural survivability of Inconel 718 keyway interlocks.

2. Submission Declarations & Publishing Model
• **Publishing Model Choice:** The author explicitly selects the **Traditional Subscription Publishing Model** (no open access author processing charges).
• **Originality:** This manuscript represents original research conducted independently by the author that has not been published previously and is not under consideration elsewhere.
• **Self-Containment:** All derivations, physical constants, equations, and literature comparisons are completely self-contained.
• **Conflicts of Interest:** The author declares no competing financial interests or personal relationships.
• **AI Disclosure:** Scientific prose, mathematical derivations, and core physics were authored directly by the investigator. Generative AI tools were used solely for code syntax optimization in data plotting scripts.

Sincerely,
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